

Persulfidation: a Proteomic Analysis

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1 Anexo III

1.1 Descriptive Analysis

1.1.1 Coefficient of Variation

The coefficient of variation was calculated as the ratio of the standard deviation to the mean of the data, and expressed as a percentage:

$$CV = (Standard\ deviation / mean) * 100$$

For this and the following analyses, the file “~/Proteins/b.csv” was used.

```
# Set the directory name
base_dir <- "/home/nicolasdi/BIXO-LINUX/-Persulfidation--A-proteomic-Analysis/"

file_path <- paste0(base_dir, "/Proteins/b.csv")
```

```

# Load the .csv file used
a <- read_csv(file_path)
a <- data.frame(a, row.names = c("C1",
                                "C2",
                                "C3",
                                "O1",
                                "O2",
                                "O3",
                                "S1",
                                "S2",
                                "S3"))

proteins <- data.frame(colnames(a)) # Extracted the names of the 1395 proteins

# From the three experimental conditions: Control ("C"), NaSH ("S"), and OGD ("O"), the CV function

# Function to calculate the mean and CV for a given subset of rows
calculate_cv_mean <- function(data, rows) {
  mean_vals <- sapply(data[rows, ], mean) # Calculate mean for the selected rows
  cv_vals <- sapply(data[rows, ], function(x) sd(x) / mean(x) * 100) # Calculate CV for the selected rows
  return(data.frame(mean = mean_vals, cv = cv_vals)) # Return as a data frame
}

# Control (CV and mean for Control samples)
dfC <- calculate_cv_mean(a, 1:3)

# NaHS (CV and mean for NaHS samples)
dfS <- calculate_cv_mean(a, 7:9)

# OGD (CV and mean for OGD samples)
dfO <- calculate_cv_mean(a, 4:6)

```

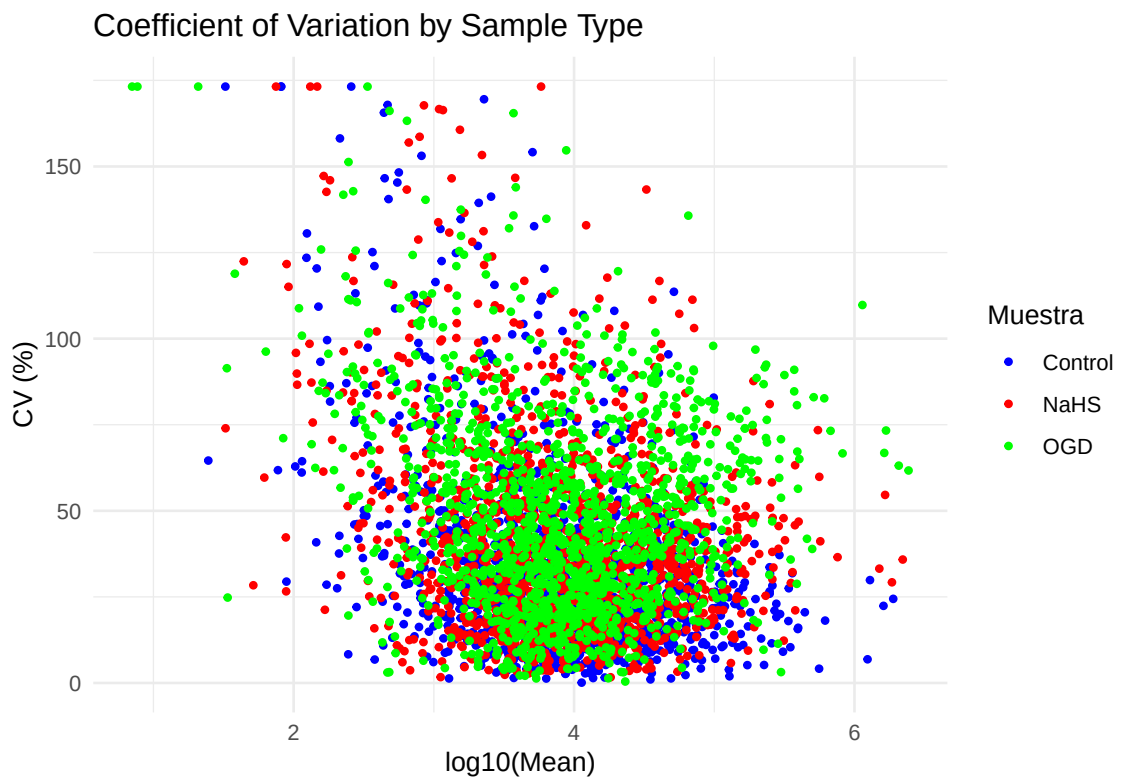
```

# Calculate log10 of the mean for each data frame explicitly
dfC <- mutate(dfC, log_x = log10(dfC$mean)) # Apply log10 to the 'mean' column in dfC
dfS <- mutate(dfS, log_x = log10(dfS$mean)) # Apply log10 to the 'mean' column in dfS
dfO <- mutate(dfO, log_x = log10(dfO$mean)) # Apply log10 to the 'mean' column in dfO

combined_data <- bind_rows(
  mutate(dfC, Muestra = "Control"),
  mutate(dfS, Muestra = "NaHS"),
  mutate(dfO, Muestra = "OGD")
)

# Plot for all samples
ggplot(combined_data, aes(x = log_x, y = cv, color = Muestra)) +
  geom_point(size = 1) +
  ggtitle("Coefficient of Variation by Sample Type") +
  xlab("log10(Mean)") + ylab("CV (%)") +
  scale_color_manual(values = c("Control" = "blue", "NaHS" = "red", "OGD" = "green")) +
  theme_minimal()

```



1.2

1.3 PCA

The PCA analysis represents the variation in the dataset expressed in transformed variables (Principal Components). The `factorextra` package is used for PCA. Data was loaded from the file “~/Proteins/b.csv” and processed as input. Data standardization was performed using a robust approach based on the median and interquartile range (IQR) for the 1395 variables (proteins) using the formula:

$$\text{Robust standardization} = (X - \text{median}) / \text{IQR}$$

```
# Calculate median and IQR for each protein (across 9 samples)
mediana <- apply(a, 2, median) # Median for each column (protein)
iqr <- apply(a, 2, IQR)       # IQR for each column (protein)

#Function for standardization
robust_scale <- function(x, mediana, iqr)
```

```

{ y=(x-mediana)/(iqr)}

#Applying robust_scale function for each protein
res <- data.frame (C1 = c(),
                   C2 = c(),
                   C3 = c(),
                   O1 = c(),
                   O2 = c(),
                   O3 = c(),
                   S1 = c(),
                   S2 = c(),
                   S3 = c())

for (j in 1:1395) {
  for (i in 1:9){
    res[i,j] <- robust_scale(a[i,j], mediana[j], iqr[j])}
  }

# Transpose the result and remove NA values
res <- data.frame(res, row.names = c("C1",
                                     "C2",
                                     "C3",
                                     "O1",
                                     "O2",
                                     "O3",
                                     "S1",
                                     "S2",
                                     "S3"))

res <- data.frame(t(res)) #Results "res" is structured for further analysis
res <- data.frame (res, row.names = t(proteins))
res <- data.frame(t(res))

```

```
res1 <- data.frame(t(na.omit(t(res))))
```

```
# PCA is performed
```

```
pca <- prcomp(res1, scale = F)
```

```
#Explained variation can be seen with summary
```

```
summary(pca)
```

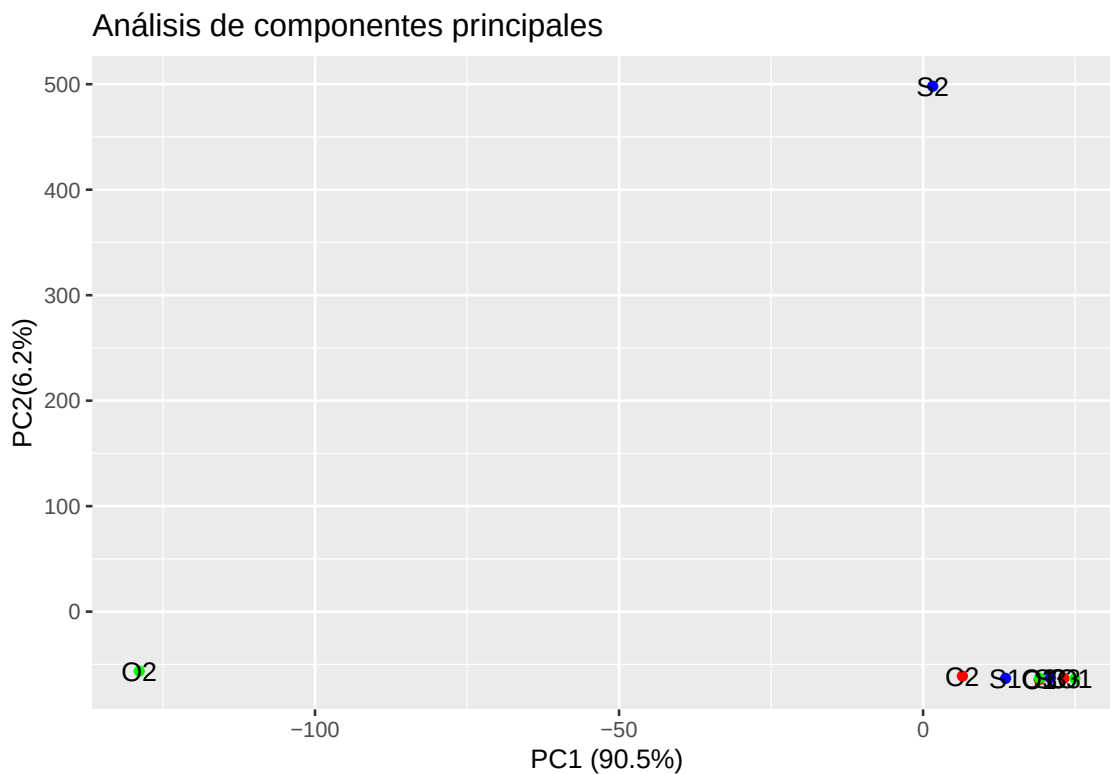
```
## Importance of components:
```

```
##              PC1      PC2      PC3      PC4      PC5      PC6
## Standard deviation 186.785 48.95787 21.90733 18.07300 13.86255 11.08348
## Proportion of Variance 0.905 0.06217 0.01245 0.00847 0.00498 0.00319
## Cumulative Proportion 0.905 0.96716 0.97961 0.98808 0.99306 0.99625
##              PC7      PC8      PC9
## Standard deviation 9.49416 7.37609 2.044e-14
## Proportion of Variance 0.00234 0.00141 0.000e+00
## Cumulative Proportion 0.99859 1.00000 1.000e+00
```

```
# Visualitazion
```

```
pcaind <- as.data.frame(pca$x) # Names were extracted
```

```
ggplot(data = pcaind, mapping = aes(y=PC1, x=PC2, z=PC3)) +
geom_point(color=c('red', 'red', 'red', 'green', 'green', 'green', 'blue', 'blue', 'blue')) +
  ggtitle("Análisis de componentes principales")+
  xlab("PC1 (90.5%)" )+ #Obtenido con la variación explicada
  ylab("PC2(6.2%)" )+
  geom_text(label=c("C1", "C2", "C3", "O1", "O2", "O3", "S1", "S2", "S3"))
```



1.4 Persulfurated protein identification

Proteins with the label identified as **DCP-Bio1** or **DCP-Bio1 hydrolyzed** were filtered from the .csv files generated in the qualitative analysis. The code for the control condition is explained in detail in 4 steps. The same procedure was applied to the other two conditions.

```
# Step 1: The .csv file of each of the 3 biological replicates was loaded as a data frame. Proteins

# Data loading

C1 <- read_csv(paste0(base_dir, "/Proteins/C1.csv"))
C2 <- read_csv(paste0(base_dir, "/Proteins/C2.csv"))
C3 <- read_csv(paste0(base_dir, "/Proteins/C3.csv"))

# Step 2: Data filtering for those with "DCP-Bio1" in the "peptide.modification" column and redu

C1filtered <- C1[grepl("DCP-Bio1", C1$peptide.modification), ]
```

```

C1filtered <- data.frame(C1filtered[, 2:4])

C2filtered <- C2[grepl("DCP-Bio1", C2$peptide.modification), ]
C2filtered <- data.frame(C2filtered[, 2:4])

C3filtered <- C3[grepl("DCP-Bio1", C3$peptide.modification), ]
C3filtered <- data.frame(C3filtered[, 2:4])

# Step 3: The 3 replicates were consolidated into a single data frame. To better identify the proteins

C1C2C3 <- rbind(C1filtered, C2filtered, C3filtered)

C1C2C3T <- data.frame(names = C1C2C3$protein.Description)

protein_list_Control <- C1

C1C2C3T <- C1C2C3T %>%
  mutate(split = strsplit(as.character(names), "=")) %>%
  transmute(
    Persulfurated_Proteins = sapply(split, `[`, 1),
    key2 = sapply(split, `[`, 2),
    key3 = sapply(split, `[`, 3),
    key4 = sapply(split, `[`, 4),
    key5 = sapply(split, `[`, 5),
    key6 = sapply(split, `[`, 6)
  )

# Step 4: A list was generated indicating the identification of persulfurated proteins.

PersulfuratedControl <- C1C2C3T %>% distinct(Persulfurated_Proteins)
PersulfuratedControl_protein.accession <- C1C2C3 %>% distinct(C1C2C3$protein.Accession)

```

```
PersulfuratedControl_protein.accession <- as.character(PersulfuratedControl_protein.accession$`C1
```

```
## NaHS
```

```
# Step 1
```

```
S1 <- read_csv(paste0(base_dir, "/Proteins/S1.csv"))
```

```
S2 <- read_csv(paste0(base_dir, "/Proteins/S2.csv"))
```

```
S3 <- read_csv(paste0(base_dir, "/Proteins/S3.csv"))
```

```
# Step 2
```

```
S1filtered <- S1[grepl("DCP-Bio1", S1$peptide.modification), ]
```

```
S1filtered <- data.frame(S1filtered[, 2:4])
```

```
S2filtered <- S2[grepl("DCP-Bio1", S2$peptide.modification), ]
```

```
S2filtered <- data.frame(S2filtered[, 2:4])
```

```
S3filtered <- S3[grepl("DCP-Bio1", S3$peptide.modification), ]
```

```
S3filtered <- data.frame(S3filtered[, 2:4])
```

```
# Step 3
```

```
S1S2S3 <- rbind(S1filtered, S2filtered, S3filtered)
```

```
S1S2S3T <- data.frame(names = S1S2S3$protein.Description)
```

```
library(tidyr)
```

```
S1S2S3T <- S1S2S3T %>%
```

```
  mutate(split = strsplit(as.character(names), "=")) %>%
```

```
  transmute(
```

```
    Persulfurated_Proteins = sapply(split, `[`, 1),
```

```
    key2 = sapply(split, `[`, 2),
```

```
    key3 = sapply(split, `[`, 3),
```

```

    key4 = sapply(split, `[`, 4),
    key5 = sapply(split, `[`, 5),
    key6 = sapply(split, `[`, 6)
  )

# Step 4

PersulfuratedNaHS <- S1S2S3T %>% distinct(Persulfurated_Proteins)
PersulfuratedNaHS_protein.accession <- S1S2S3 %>% distinct(S1S2S3$protein.Accession)
PersulfuratedNaHS_protein.accession <- as.character(PersulfuratedNaHS_protein.accession$`S1S2S3$protein.Accession`)

# OGD

# Step 1
O1 <- read_csv(paste0(base_dir, "/Proteins/O1.csv"))
O2 <- read_csv(paste0(base_dir, "/Proteins/O2.csv"))
O3 <- read_csv(paste0(base_dir, "/Proteins/O3.csv"))

# Step 2

O1filtered <- O1[grepl("DCP-Bio1", O1$peptide.modification), ]
O1filtered <- data.frame(O1filtered[, 2:4])

O2filtered <- O2[grepl("DCP-Bio1", O2$peptide.modification), ]
O2filtered <- data.frame(O2filtered[, 2:4])

O3filtered <- O3[grepl("DCP-Bio1", O3$peptide.modification), ]
O3filtered <- data.frame(O3filtered[, 2:4])

# Step 3

O1O2O3 <- rbind(O1filtered, O3filtered, O2filtered)

```

```

010203T <- data.frame(names = 010203$protein.Description)

library(tidyr)
010203T <- 010203T %>%
  mutate(split = strsplit(as.character(names), "=") %>%
    transmute(
      Persulfurated_Proteins = sapply(split, `[`, 1),
      key2 = sapply(split, `[`, 2),
      key3 = sapply(split, `[`, 3),
      key4 = sapply(split, `[`, 4),
      key5 = sapply(split, `[`, 5),
      key6 = sapply(split, `[`, 6)
    )

# Step 4

PersulfuratedOGD <- 010203T %>% distinct(Persulfurated_Proteins)
PersulfuratedOGD_protein.accession <- 010203 %>% distinct(010203$protein.Accession)
PersulfuratedOGD_protein.accession <- as.character(PersulfuratedOGD_protein.accession`010203$pro

```

1.4.1 Protein visualization

```

# Transformation of data frames into vectors
PersulfuratedControl <- as.character(PersulfuratedControl$Persulfurated_Proteins)
PersulfuratedNaHS <- as.character(PersulfuratedNaHS$Persulfurated_Proteins)
PersulfuratedOGD <- as.character(PersulfuratedOGD$Persulfurated_Proteins)

# Creation of the Venn diagram
venn.plot <- venn.diagram(
  x = list(Set1 = PersulfuratedControl,
           Set2 = PersulfuratedNaHS,
           Set3 = PersulfuratedOGD),

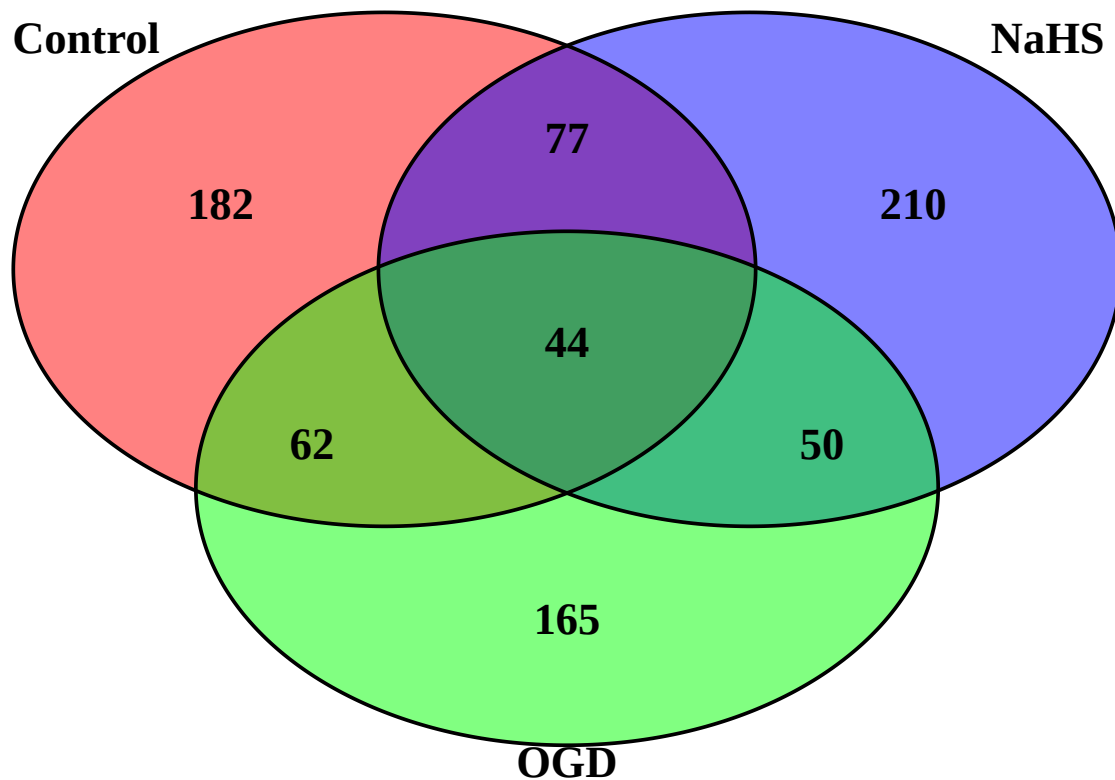
```

```

category.names = c("Control", "NaHS", "OGD"),
fill = c("red", "blue", "green"), # Circle colors
alpha = 0.5, # Transparency of the colors
cat.cex = 1.5, # Category text size
cex = 1.5, # Number text size
fontface = "bold", # Text style
cat.fontface = "bold",
filename = NULL)

# Draw the diagram
grid.draw(venn.plot)

```



1.4.1.1 Control

```

# Proteins identified only in the Control condition
Venn_Control <- setdiff(PersulfuratedControl, union(PersulfuratedNaHS, PersulfuratedOGD))
Lista_Control <- setdiff(PersulfuratedControl_protein.accession, union(PersulfuratedNaHS_protein.

```

```
knitr::kable(sort(Venn_Control))
```

x

[histone H3]-trimethyl-L-lysine(9) demethylase OS
[tau protein] kinase OS
26S proteasome non-ATPase regulatory subunit 13 OS
26S proteasome non-ATPase regulatory subunit 2 OS
26S proteasome regulatory subunit 6A OS
Actin related protein 2/3 complex__ subunit 1B OS
Actin-related protein 2/3 complex subunit 1B OS
ADP/ATP translocase 2 OS
Albumin OS
Aldehyde dehydrogenase__ cytosolic 1 OS
Aldehyde oxidase 4 OS
aldehyde oxidase OS
Alkaline phosphatase__ tissue-nonspecific isozyme OS
All-trans-retinol 13_14-reductase OS
Annexin A1 OS
ATPase family gene 2 protein homolog B OS
Biliverdin reductase A OS
BTB domain and CNC homolog 2 OS
Calreticulin OS
cAMP-dependent protein kinase catalytic subunit beta OS
cAMP-dependent protein kinase OS
Centrosomal protein 290 OS
Citrate synthase OS
Citrate synthase__ mitochondrial OS
Coiled coil domain containing 88A OS
Coiled-coil domain containing 180 OS
COP9 signalosome complex subunit 1 OS
COP9 signalosome complex subunit 4 OS
Copine 5 OS
Copine family member 9 OS

Copine-9 OS
Creatine kinase B-type OS
creatine kinase OS
Cytoplasmic linker associated protein 1 OS
D-3-phosphoglycerate dehydrogenase OS
DEAD-box helicase 3_ Y-linked OS
Delta(3_5)-Delta(2_4)-dienoyl-CoA isomerase__ mitochondrial OS
Dihydropyrimidinase-related protein 2 OS
DNA polymerase nu OS
DNA-(apurinic or apyrimidinic site) endonuclease OS
Doublecortin-like kinase 1 OS
Dynein axonemal heavy chain 7 OS
Dynein__ axonemal__ heavy chain 7 OS
Elongation factor 1-gamma OS
Enoyl-CoA hydratase 1 OS
Eukaryotic translation elongation factor 1 gamma OS
Eukaryotic translation initiation factor 2 subunit beta OS
Exportin 1 OS
Exportin-1 OS
FA complementation group F OS
Far upstream element-binding protein 2 OS
FAT atypical cadherin 4 OS
Fer-1-like family member 6 OS
FERM domain containing kindlin 2 OS
Fructose-bisphosphate aldolase A OS
Fructose-bisphosphate aldolase C OS
fructose-bisphosphate aldolase OS
Fructose-bisphosphate aldolase OS
G protein pathway suppressor 1 OS
G protein subunit alpha 14 OS
Glutamate-cysteine ligase OS

Glutathione S-transferase Mu 1 OS
 Glutathione S-transferase mu 2 OS
 Glutathione S-transferase Mu 2 OS
 Glutathione S-transferase Mu 4 OS
 Glutathione S-transferase Mu 5 OS
 Glutathione S-transferase Mu 7 OS
 Glutathione S-transferase P OS
 glutathione transferase OS
 Glycerol-3-phosphate dehydrogenase OS
 Glycerol-3-phosphate dehydrogenase__ mitochondrial OS
 Glycogen synthase kinase-3 beta OS
 Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-3 OS
 Guanylate binding protein 2 OS
 Guanylate-binding protein 2 OS
 Heat shock protein 105 kDa OS
 Heat shock protein family H (Hsp110) member 1 OS
 Heterogeneous nuclear ribonucleoprotein L OS
 HORMA domain containing 2 OS
 Huntingtin OS
 Importin subunit beta-1 OS
 inorganic diphosphatase OS
 Isoaspartyl peptidase/L-asparaginase OS
 Isoform 2 of Gelsolin OS
 Isoform 2 of Mitogen-activated protein kinase 3 OS
 Isoform 2 of NADH-cytochrome b5 reductase 3 OS
 Isoform 2 of Septin-11 OS
 Isoform 2 of Voltage-dependent anion-selective channel protein 3 OS
 Isoform 3 of NADH-cytochrome b5 reductase 3 OS
 Isoform 3 of Septin-11 OS
 Isoform Alpha-2 of Dual specificity mitogen-activated protein kinase kinase 5 OS
 Karyopherin subunit beta 1 OS

Keratin 2 OS
Keratin 31 OS
Keratin 33B OS
Keratin 76 OS
Keratin 85 OS
Keratin__ type I cytoskeletal 13 OS
Keratin__ type I cytoskeletal 40 OS
Keratin__ type II cytoskeletal 2 epidermal OS
KH-type splicing regulatory protein OS
L-lactate dehydrogenase A chain OS
Large ribosomal subunit protein uL3 OS
Large ribosomal subunit protein uL4 OS
Ligand dependent nuclear receptor corepressor-like OS
Lysine demethylase 4C OS
MAP kinase-activated protein kinase 3 OS
MAPK activated protein kinase 3 OS
Mitogen activated protein kinase kinase kinase 4 OS
Mitogen-activated protein kinase 1 OS
Mitogen-activated protein kinase 3 OS
Mitogen-activated protein kinase 4 (Fragment) OS
Mitogen-activated protein kinase OS
Myomesin 2 OS
Myosin__ heavy chain 9 OS
N-acyl phosphatidylethanolamine phospholipase D OS
NADH-cytochrome b5 reductase 3 OS
NADH-cytochrome b5 reductase OS
Nischarin OS
Nuclear pore complex protein Nup93 OS
Nurim OS
Ornithine aminotransferase__ mitochondrial OS
Phosphatidylinositol-3-phosphate phosphatase OS

Plectin OS
Poly(rC) binding protein 1 OS
POU domain__ class 5__ transcription factor 2 OS
Proteasome (Prosome__ macropain) 26S subunit__ ATPase 3 OS
Proteasome 26S subunit__ ATPase 3 OS
Protein tyrosine kinase 7 OS
Protein tyrosine phosphatase__ non-receptor type 9 OS
Protein tyrosine phosphatase__ receptor type N2 OS
protein-tyrosine-phosphatase OS
Rab proteins geranylgeranyltransferase component A OS
RAD51 associated protein 2 OS
Ras-related protein Rab-4 OS
Ras-related protein Rab-4A OS
Receptor-type tyrosine-protein phosphatase N2 OS
Replication timing regulatory factor 1 OS
Retinol saturase OS
Retinol-binding protein 1 OS
Reversed Sequence 13150
Reversed Sequence 204
Reversed Sequence 2112
Reversed Sequence 28989
Reversed Sequence 4435
Reversed Sequence 5304
Ribosomal protein L3 OS
Ring finger protein 213 OS
RRM domain-containing protein OS
Sept4 protein OS
Septin-11 OS
Septin-4 OS
Serine and arginine rich splicing factor 3 OS
Serine/threonine-protein phosphatase 2A 55 kDa regulatory subunit B alpha isoform OS

Serine/threonine-protein phosphatase 2A 55 kDa regulatory subunit B OS
 Similar to 60S ribosomal protein L4 (L1) OS
 Similar to Lung carbonyl reductase [NADPH] (NADPH-dependent carbonyl reductase) (LCR)
 (Adipocyte P27 protein) (AP27) OS
 Similar to OTTMUSP00000000621 OS
 Solute carrier family 25 member 3 OS
 Somatomedin B and thrombospondin__ type 1 domain containing OS
 Spermatogenesis associated 5-like 1 OS
 Splicing factor 3b__ subunit 1 OS
 Splicing factor proline and glutamine rich OS
 Synaptotagmin-like 2 OS
 T-box brain transcription factor 1 OS
 T-complex protein 1 subunit theta OS
 Telomerase associated protein 1 OS
 Telomerase protein component 1 OS
 Thioredoxin reductase 1__ cytoplasmic OS
 thioredoxin-disulfide reductase OS
 Transmembrane protein 54 OS
 Tubulin beta-5 chain OS
 Twinfilin-1 OS
 Tyrosine-protein phosphatase non-receptor type 9 OS
 ubiquitinyl hydrolase 1 OS
 V-type proton ATPase subunit B__ brain isoform OS
 Vacuolar protein sorting-associated protein 35 OS
 Vacuolar proton pump subunit B OS
 Voltage-dependent anion channel 3 OS
 Voltage-dependent anion-selective channel protein 3 OS
 Zinc finger protein 431-like OS
 Zinc finger protein 729-like OS

1.4.1.2 Control and NaHS

```
# Proteins identified only in the Control and NaHS condition
```

```
Venn_Control_X_NaHS <- setdiff(intersect(PersulfuratedControl, PersulfuratedNaHS), PersulfuratedNaHS)
```

```
Lista_Control_X_NaHS <- setdiff(intersect(PersulfuratedControl_protein.accession, PersulfuratedNaHS_protein.accession), PersulfuratedNaHS_protein.accession)
```

```
knitr::kable(Venn_Control_X_NaHS)
```

x

Alpha-enolase OS

Peptidyl-prolyl cis-trans isomerase A OS

Peptidyl-prolyl cis-trans isomerase OS

Elongation factor 1-alpha OS

Elongation factor 1-alpha 1 OS

Eukaryotic translation elongation factor 1 alpha 1 OS

Tr-type G domain-containing protein OS

Transitional endoplasmic reticulum ATPase OS

Acyl-CoA dehydrogenase family__ member 10 OS

Small ribosomal subunit protein RACK1 OS

Keratin 71 OS

Heterogeneous nuclear ribonucleoprotein M OS

Myosin-9 OS

Proteasome subunit beta type-2 OS

ceramidase OS

Acid ceramidase OS

PDZ and LIM domain 5 OS

PDZ and LIM domain protein 5 OS

Isoform 2 of PDZ and LIM domain protein 5 OS

Myosin heavy chain 10 OS

Myosin-10 OS

Isoform 2 of Dihydropyrimidinase-related protein 3 OS

Dihydropyrimidinase-like 3 OS

Dihydropyrimidinase-related protein 3 OS

Centromere protein F OS

Heterogeneous nuclear ribonucleoprotein U OS

Sorbin and SH3 domain containing 1 OS

Coatomer subunit gamma OS
Coatomer subunit gamma-1 OS
Vacuolar protein sorting 13 homolog D OS
Zinc finger FYVE-type containing 1 OS
Kinesin family member 20B OS
AP-2 complex subunit alpha-2 OS
Glutathione S-transferase OS
G protein subunit alpha 13 OS
Guanine nucleotide-binding protein subunit alpha-13 OS
Aldehyde dehydrogenase 1 family__ member A1 OS
Aldehyde dehydrogenase 1A1 OS
Protein phosphatase 2 scaffold subunit A alpha OS
N(G)_N(G)-dimethylarginine dimethylaminohydrolase 2 OS
Glutamine synthetase OS
Isoform 2 of Dynein axonemal heavy chain 1 OS
Dynein__ axonemal__ heavy chain 1 OS
Dynein axonemal heavy chain 1 OS
Spliceosome RNA helicase Ddx39b OS
DEXD-box helicase 39B OS
Septin 4 OS
Eukaryotic translation initiation factor 4__ gamma 2 OS
Cofilin-1 OS
Cofilin 1 OS
L-lactate dehydrogenase OS
Eukaryotic translation initiation factor 5A OS
Eukaryotic translation initiation factor 5A-1 OS
Eukaryotic translation initiation factor 4A2 OS
Glutaredoxin 3 OS
Glutaredoxin-3 OS
Actin-related protein 3 OS
Actin related protein 3 OS

x

Protein phosphatase 2 scaffold subunit A beta OS
Serine/threonine-protein phosphatase 2A 65 kDa regulatory subunit A beta isoform OS
Stress-70 protein__ mitochondrial OS
Staphylococcal nuclease domain-containing protein 1 OS
Staphylococcal nuclease domain-containing protein OS
Staphylococcal nuclease and tudor domain containing 1 OS
Lamin B1 OS
Malic enzyme OS
NADP-dependent malic enzyme OS
Ciliary rootlet coiled-coil__ rootletin OS
Adenylate kinase 9 OS
Transcription activation suppressor family member 2 OS
Lamin-B1 OS
Senataxin OS
Uncharacterized protein OS
non-specific serine/threonine protein kinase OS
Hemicentin 2 OS
Dynein__ axonemal__ heavy chain 17 OS
Glutamate-cysteine ligase catalytic subunit OS

1.4.1.3 Control and OGD

```
# Proteins identified only in the Control and OGD condition
```

```
Venn_Control_X_OGD <- setdiff(intersect(PersulfuratedControl, PersulfuratedOGD), PersulfuratedNaH  
Lista_Control_X_OGD <- setdiff(intersect(PersulfuratedControl_protein.accession, PersulfuratedOGD  
knitr::kable(sort(Venn_Control_X_OGD))
```

x

Actinin alpha 3 OS
Actinin__ alpha 1 OS
ADP/ATP translocase OS
Alpha-actinin-1 OS

Biorientation of chromosomes in cell division 1-like 1 OS
Centromere protein E OS
Copine 8 OS
Cytoplasmic dynein 2 heavy chain 1 OS
DEAD box helicase 17 OS
Dihydropyrimidinase-like 2 OS
Dynein cytoplasmic 2 heavy chain 1 OS
E1 ubiquitin-activating enzyme OS
Echinoderm microtubule-associated protein-like 2 OS
Electron transfer flavoprotein subunit alpha OS
Electron transfer flavoprotein subunit alpha__ mitochondrial OS
EMAP like 2 OS
enoyl-CoA hydratase OS
Fibronectin OS
Filamin C OS
Filamin-C OS
Four and a half LIM domains 1 OS
Four and a half LIM domains protein 1 OS
Galectin-1 OS
glutamate dehydrogenase [NAD(P)(+)] OS
Glutamate dehydrogenase 1__ mitochondrial OS
GNAS complex locus OS
Guanine nucleotide-binding protein G(s) subunit alpha isoforms XLas OS
Heat shock 70 kDa protein 4 OS
Heat shock protein family A (Hsp70) member 4 OS
Hypoxanthine phosphoribosyltransferase OS
Hypoxanthine-guanine phosphoribosyltransferase OS
Isoform 2 of Echinoderm microtubule-associated protein-like 2 OS
Isoform 2 of Fibronectin OS
Isoform 3 of Fibronectin OS
Isoform 4 of Fibronectin OS

x

Isoform M2 of Pyruvate kinase PKM OS

Keratin 36 OS

Keratin_ type II cytoskeletal 1 OS

Moesin OS

Myosin heavy chain 14 OS

NADH dehydrogenase [ubiquinone] 1 alpha subcomplex subunit 10_ mitochondrial OS

Peroxiredoxin-6 OS

Plastin 3 (T-isoform) OS

Plastin-3 OS

Protein-glutamine gamma-glutamyltransferase 2 OS

protein-glutamine gamma-glutamyltransferase OS

Pyruvate kinase OS

Radixin OS

S-phase kinase associated protein 2 OS

Succinate dehydrogenase [ubiquinone] flavoprotein subunit_ mitochondrial OS

Sulfide quinone oxidoreductase OS

T-complex protein 1 subunit zeta OS

Thioredoxin domain-containing protein OS

Transgelin OS

Transgelin-2 OS

Transgelin-3 OS

Trifunctional enzyme subunit alpha_ mitochondrial OS

Tubulin beta chain OS

Tubulin beta-2A chain OS

Tubulin beta-2B chain OS

Tyrosine-protein kinase OS

von Willebrand factor A domain-containing protein 5A OS

1.4.1.4 Control, NaHS y OGD

```
# Proteins identified only in the Control, NaHS, and OGD condition
```

```
Venn_Control_X_OGD_X_NaHS <- Reduce(intersect, list(PersulfuratedControl, PersulfuratedNaHS, PersulfuratedOGD))
```

```
Lista_Control_X_OGD_NaHS <- Reduce(intersect, list(PersulfuratedControl_protein.accession, PersulfuratedNaHS_protein.accession, PersulfuratedOGD_protein.accession))
```

```
knitr::kable(sort(Venn_Control_X_OGD_X_NaHS))
```

x

Actin__ beta-like 2 OS

Annexin OS

AP-2 complex subunit alpha OS

ATP-dependent RNA helicase OS

Calponin OS

Catenin alpha 1 OS

Clathrin heavy chain 1 OS

Clathrin heavy chain OS

Cysteine and glycine-rich protein 1 OS

Dynein axonemal heavy chain 5 (Fragment) OS

Dynein__ axonemal__ heavy chain 5 OS

Eukaryotic initiation factor 4A-II OS

Filamin A OS

Filamin B OS

Gelsolin OS

Isocitrate dehydrogenase [NADP] OS

Isocitrate dehydrogenase [NADP]__ mitochondrial OS

Keratin 35 OS

Keratin 73 OS

Keratin__ type II cytoskeletal 73 OS

Mitochondrial carrier 2 OS

NADH-ubiquinone oxidoreductase 75 kDa subunit__ mitochondrial OS

Niban apoptosis regulator 2 OS

Palladin-like 1 OS

Peroxiredoxin 3 OS

phosphopyruvate hydratase OS

Protein Niban 2 OS

x

Ribonuclease inhibitor OS
Ribonuclease/angiogenin inhibitor 1 OS
RNA helicase OS
Septin OS
Sodium/potassium-transporting ATPase subunit alpha OS
Sodium/potassium-transporting ATPase subunit alpha-3 OS
T-complex protein 1 subunit epsilon OS
T-complex protein 1 subunit eta OS
T-complex protein 1 subunit gamma OS
Talin 1 OS
Thioredoxin-dependent peroxide reductase__ mitochondrial OS
Triosephosphate isomerase OS
Tubulin alpha chain OS
Tubulin alpha-4A chain OS
Ubiquitin-like modifier-activating enzyme 1 OS
von Willebrand factor A domain containing 5A OS
WD repeat-containing protein 1 OS

1.4.1.5 NaHS

Proteins identified only in the NaHS condition

```
Venn_NaHS <- setdiff(PersulfuratedNaHS, union(PersulfuratedControl, PersulfuratedOGD))  
Lista_NaHS <- setdiff(PersulfuratedNaHS_protein.accession, union(PersulfuratedControl_protein.accession, PersulfuratedOGD_protein.accession))  
knitr::kable(sort(Venn_NaHS))
```

x

2'_3'-cyclic-nucleotide 3'-phosphodiesterase OS
26S proteasome non-ATPase regulatory subunit 11 OS
3-ketoacyl-CoA thiolase A_ peroxisomal OS
3-ketoacyl-CoA thiolase B_ peroxisomal OS
4-aminobutyrate aminotransferase OS
4-aminobutyrate aminotransferase__ mitochondrial OS

4-trimethylaminobutyraldehyde dehydrogenase OS
 40S ribosomal protein S26 OS
 40S ribosomal protein S6 OS
 40S ribosomal protein SA OS
 6-phosphofructokinase OS
 6-phosphogluconolactonase OS
 60 kDa heat shock protein__ mitochondrial OS
 A-kinase anchoring protein 10 OS
 Ac2-032 OS
 Acetyl-CoA acyltransferase 1A OS
 Acetyl-Coenzyme A acyltransferase 1B OS
 Acid-sensing ion channel 4 OS
 Actin__ alpha cardiac muscle 1 OS
 Actin__ alpha skeletal muscle OS
 Actin__ alpha__ cardiac muscle 1 OS
 Actin__ aortic smooth muscle OS
 Actin__ gamma-enteric smooth muscle OS
 Actin-related protein 2/3 complex subunit 2 OS
 Adenosylhomocysteinase-like 1 OS
 ADF-H domain-containing protein OS
 Aldehyde dehydrogenase 9 family__ member A1 OS
 Aldehyde dehydrogenase__ mitochondrial OS
 Aldo-keto reductase family 1 member A1 OS
 Alpha-1__4 glucan phosphorylase OS
 Amyloid beta precursor protein binding family B member 1 OS
 Ankyrin repeat and SAM domain-containing protein 6 OS
 ArfGAP with SH3 domain__ ankyrin repeat and PH domain 2 OS
 Arp2/3 complex 34 kDa subunit OS
 asparagine-tRNA ligase OS
 ATP binding cassette subfamily A member 8 OS
 ATP synthase F(0) complex subunit B1__ mitochondrial OS

ATP synthase subunit b OS
ATP-citrate synthase OS
ATP-dependent 6-phosphofructokinase OS
ATP-dependent 6-phosphofructokinase__ muscle type OS
ATP-dependent 6-phosphofructokinase__ platelet type OS
Biogenesis of lysosome-related organelles complex 1 subunit 1 OS
C2 domain containing 3 centriole elongation regulator OS
Calpain 2 OS
Calpain-2 catalytic subunit OS
Carbonyl reductase (NADPH) OS
Carbonyl reductase [NADPH] 1 OS
Catenin alpha 3 OS
Catenin beta-1 OS
Caveolin OS
Caveolin-1 OS
Centrosomal protein of 162 kDa OS
Core histone macro-H2A OS
Core histone macro-H2A.1 OS
Cyclin-dependent kinase 6 OS
Cysteine-rich protein 2 OS
Dehydrogenase/reductase 1 OS
Destrin OS
Dipeptidyl peptidase 2 OS
Dipeptidylpeptidase 7 OS
Dynein__ axonemal__ heavy chain 9 OS
Dystonin OS
Elongation factor 1-alpha 2 OS
Elongation factor 2 OS
Epithelial cell transforming 2 OS
Epoxide hydrolase 1 OS
Epoxide hydrolase OS

Eukaryotic translation elongation factor 2 OS
Family with sequence similarity 193__ member B OS
Fatty acid binding protein 9 OS
Fatty acid-binding protein 9 OS
Fer-1-like family member 4 OS
Gamma-enolase OS
Glutamyl-prolyl-tRNA synthetase OS
Glyceraldehyde 3-phosphate dehydrogenase NAD(P) binding domain-containing protein OS
Glycogen phosphorylase__ brain form (Fragment) OS
Glycogen phosphorylase__ muscle form OS
GTP-binding nuclear protein Ran OS
GTP-binding nuclear protein Ran__ testis-specific isoform OS
GTPase activating protein testicular GAP1 OS
Heat shock 70 kDa protein 1A OS
Heat shock 70 kDa protein 1B OS
Heat shock-related 70 kDa protein 2 OS
Helicase for meiosis 1 OS
Heterogeneous nuclear ribonucleoprotein R OS
Heterogeneous nuclear ribonucleoprotein R-like OS
Hexokinase-1 OS
Huntingtin interacting protein 1 OS
Hydroxysteroid (17-beta) dehydrogenase 10 OS
IF rod domain-containing protein OS
IKAROS family zinc finger 4 OS
Isoform 2 of Amyloid beta precursor protein binding family B member 1 OS
Isoform 2 of ATP-citrate synthase OS
Isoform 2 of Caveolin-1 OS
Isoform 2 of Core histone macro-H2A.1 OS
Isoform 2 of Kalirin OS
Isoform 2 of Septin-5 OS
Isoform 3 of Kalirin OS

Isoform 3 of Septin-5 OS
Isoform 4 of Kalirin OS
Isoform 7 of Kalirin OS
Isoform 7 of SH3 domain-containing kinase-binding protein 1 OS
Isoform Gnas-3 of Guanine nucleotide-binding protein G(s) subunit alpha isoforms short OS
Junction plakoglobin OS
Kalirin OS
Keratin 6A OS
Keratin__ type I cytoskeletal 24 OS
Keratin__ type II cytoskeletal 6A OS
Keratin__ type II cytoskeletal 75 OS
Kinesin family member 15 OS
Kinesin family member 21B OS
Kinesin-like protein KIF15 OS
Kinesin-like protein KIF21B OS
L-lactate dehydrogenase C chain OS
L-threonine dehydrogenase OS
Large ribosomal subunit protein eL14 OS
Leucine rich repeat containing 59 OS
Leucine-rich repeat-containing protein 59 OS
Lysine-tRNA ligase OS
Malate dehydrogenase__ cytoplasmic OS
Microtubule-associated protein OS
Myosin IE OS
Myosin IF OS
Nidogen-2 OS
Ninein-like OS
Nuclear receptor co-repressor 1 OS
Palladin (Fragment) OS
PDZ and LIM domain 4 OS
PDZ and LIM domain protein 4 OS

Pecanex-like protein OS
 Peroxiredoxin C-terminal domain-containing protein OS
 Peroxiredoxin-1 OS
 Phenylalanine-tRNA ligase beta subunit OS
 phenylalanine-tRNA ligase OS
 Phosphoglucomutase 1 OS
 Phosphoglucomutase-1 OS
 Phosphoglycerate kinase OS
 Phosphoglycerate mutase 1 OS
 Phospholipid-transporting ATPase OS
 Poly (ADP-ribose) polymerase family__ member 3 OS
 Poly [ADP-ribose] polymerase OS
 Pre-mRNA-processing factor 6 OS
 proline-tRNA ligase OS
 Prostaglandin E synthase 3 OS
 Proteasome 26S subunit__ non-ATPase 11 OS
 Proteasome subunit beta OS
 Proteasome subunit beta type-1 OS
 Proteasome subunit beta type-6 OS
 Pyruvate dehydrogenase E1 component subunit alpha OS
 Pyruvate dehydrogenase E1 component subunit alpha__ somatic form__ mitochondrial OS
 Rab GDP dissociation inhibitor alpha OS
 Rab GDP dissociation inhibitor beta OS
 Rab GDP dissociation inhibitor OS
 RAB2A__ member RAS oncogene family OS
 RAB2B__ member RAS oncogene family OS
 RAB33B__ member RAS oncogene family OS
 Ras-related protein Rab-2A OS
 RB1-inducible coiled-coil 1 OS
 Retinaldehyde binding protein 1 OS
 Reversed Sequence 2732

Reversed Sequence 28683
Reversed Sequence 3489
Reversed Sequence 35088
Reversed Sequence 37029
Reversed Sequence 39294
Reversed Sequence 41012
Reversed Sequence 4678
Reversed Sequence 594
Reversed Sequence 7146
Reversed Sequence 7827
Rho guanine nucleotide exchange factor 2 OS
Rho/Rac guanine nucleotide exchange factor 18 OS
Rho/Rac guanine nucleotide exchange factor 2 OS
RLF zinc finger OS
RP1__ axonemal microtubule associated OS
Ryanodine receptor 1 OS
S-adenosylhomocysteine hydrolase-like protein 1 OS
Schlafen family member 13 OS
Sepiapterin reductase OS
Septin-5 OS
Serine/threonine-protein phosphatase 2A catalytic subunit alpha isoform OS
Serine/threonine-protein phosphatase 2A catalytic subunit beta isoform OS
Serine/threonine-protein phosphatase 2B catalytic subunit beta isoform OS
Signal peptidase complex subunit 3 OS
Similar to FLJ44112 protein OS
Small ribosomal subunit protein eS26 OS
Small ribosomal subunit protein eS6 OS
Small ribosomal subunit protein uS17 N-terminal domain-containing protein OS
Small ribosomal subunit protein uS17 OS
Small ribosomal subunit protein uS2 OS
Small ribosomal subunit protein uS5 C-terminal domain-containing protein OS

x

Small ribosomal subunit protein uS5 OS
Sperm associated antigen 17 OS
Sphingosine-1-phosphate phosphatase 1 OS
Succinate dehydrogenase [ubiquinone] iron-sulfur subunit__ mitochondrial OS
Suppressor of glucose__ autophagy associated 1 OS
Synaptic vesicle membrane protein VAT-1 homolog OS
T-complex protein 1 subunit beta OS
T-complex protein 1 subunit delta OS
TBC1 domain family__ member 4 OS
Tetratricopeptide repeat domain 8 OS
Type II keratin Kb15 OS
Ubiquitin carboxyl-terminal hydrolase OS
Unconventional myosin-Ie OS
Utrophin OS
Vacuolar protein-sorting-associated protein 25 OS
Very-long-chain 3-oxoacyl-CoA reductase OS
Zinc finger protein 518A OS
Zyxin OS

1.4.1.6 OGD

Proteins identified only in the OGD condition

```
Venn_OGD <- setdiff(PersulfuratedOGD, union(PersulfuratedControl, PersulfuratedNaHS))  
Lista_OGD <- setdiff(PersulfuratedOGD_protein.accession, union(PersulfuratedControl_protein.accession, PersulfuratedNaHS_protein.accession))  
knitr::kable(sort(Venn_OGD))
```

x

1-phosphatidylinositol 4_5-bisphosphate phosphodiesterase delta-1 OS
2-iminobutanoate/2-iminopropanoate deaminase OS
40S ribosomal protein S15a OS
6-phosphogluconate dehydrogenase__ decarboxylating OS
Ac1233 OS

Acetyl-CoA acetyltransferase__ mitochondrial OS
Actinin alpha 2 OS
Actinin alpha 4 OS
Adenosine 5'-monophosphoramidase HINT1 OS
Adenosylhomocysteinase OS
ADP/ATP translocase 1 OS
Aldo-keto reductase family 1 member B OS
Aldo-keto reductase family 1 member B1 OS
Alpha-actinin-4 OS
Aminopeptidase OS
Aminopeptidase puromycin sensitive OS
Annexin A2 OS
Annexin A4 OS
arachidonate-CoA ligase OS
Aspartate aminotransferase OS
Aspartate aminotransferase__ mitochondrial OS
Aspartate-tRNA ligase__ cytoplasmic OS
ATP-dependent 6-phosphofructokinase__ liver type OS
B-cell receptor-associated protein OS
BAG cochaperone 2 OS
Beta-adaptin appendage C-terminal subdomain domain-containing protein OS
Beta-soluble NSF attachment protein OS
Calcium regulated heat stable protein 1 OS
Calcium-regulated heat stable protein 1 OS
Cathepsin D OS
Centriolin OS
Chaperonin containing TCP1 subunit 6A OS
Coatmer subunit delta OS
Copine 2 OS
DEAD-box helicase 6 OS
Delta-1-pyrroline-5-carboxylate dehydrogenase__ mitochondrial OS

Delta-1-pyrroline-5-carboxylate synthase OS
Dihydrolipoyl dehydrogenase OS
Dihydrolipoyl dehydrogenase__ mitochondrial OS
DNA damage-binding protein 1 OS
DNA polymerase delta interacting protein 2 OS
DNA polymerase epsilon catalytic subunit OS
E3 ubiquitin-protein ligase TTC3 OS
Elastin microfibril interfacier 1 OS
Endoplasmic reticulum aminopeptidase 1 OS
Endoplasmic reticulum resident protein 29 OS
ERCC excision repair 6 like 2 OS
Erythrocyte membrane protein band 4.1-like 2 OS
Eukaryotic translation initiation factor 3 subunit D OS
Eukaryotic translation initiation factor 3 subunit I OS
Fibroblast growth factor receptor OS
G protein subunit alpha o1 OS
G protein subunit alpha q OS
GC-rich promoter binding protein 1-like 2 OS
GDP-L-fucose synthase OS
Guanine nucleotide-binding protein G(q) subunit alpha OS
Heat shock cognate 71 kDa protein OS
Heat shock cognate protein 70__ pseudogene 1 OS
Heat shock protein family A (Hsp70) member 8 OS
High density lipoprotein binding protein OS
Histidine triad nucleotide binding protein 1__ pseudogene 1 OS
Histone deacetylase 2 OS
Hydroxysteroid (17-beta) dehydrogenase 4 OS
Importin 4 OS
Importin 5 OS
Isoform 10a of Neurexin-1 OS
Isoform 11a of Neurexin-1 OS

Isoform 12a of Neurexin-1 OS
 Isoform 13a of Neurexin-1 OS
 Isoform 2 of Protein phosphatase 3 catalytic subunit alpha OS
 Isoform 2 of SERPINE1 mRNA-binding protein 1 OS
 Isoform 2a of Neurexin-1 OS
 Isoform 3 of Protein piccolo OS
 Isoform 3a of Neurexin-1 OS
 Isoform 4a of Neurexin-1 OS
 Isoform 5a of Neurexin-1 OS
 Isoform 6a of Neurexin-1 OS
 Isoform 7a of Neurexin-1 OS
 Isoform 8a of Neurexin-1 OS
 Isoform 9a of Neurexin-1 OS
 Isoform Alpha-2 of Guanine nucleotide-binding protein G(o) subunit alpha OS
 Isoform Gnas-2 of Guanine nucleotide-binding protein G(s) subunit alpha isoforms short OS
 Isoform Long of Annexin A2 OS
 Isoform Short of Potassium-transporting ATPase alpha chain 2 OS
 K(2P) channel-a subtype OS
 Keratin 25 OS
 Keratin 78 OS
 Keratin_ type I cytoskeletal 25 OS
 Large ribosomal subunit protein uL5 OS
 Leucine-rich pentatricopeptide repeat containing OS
 Leucine-rich PPR motif-containing protein_ mitochondrial OS
 Leucine-rich repeats and guanylate kinase domain containing OS
 Long-chain-fatty-acid-CoA ligase 5 OS
 Long-chain-fatty-acid-CoA ligase OS
 Malate dehydrogenase_ mitochondrial OS
 Mitochondrial proton/calcium exchanger protein OS
 Mitogen-activated protein kinase 6 OS
 Multifunctional fusion protein OS

Multifunctional protein ADE2-like OS
 Na(+)/H(+) exchange regulatory cofactor NHE-RF1 OS
 NADH dehydrogenase [ubiquinone] 1 beta subcomplex subunit 10 OS
 Neurexin 1 OS
 Neurexin-1 OS
 non-specific protein-tyrosine kinase OS
 NSF attachment protein beta OS
 Nucleic acid binding protein OS
 Peptidyl-prolyl cis-trans isomerase D OS
 Peptidylprolyl isomerase D OS
 Peptidylprolyl isomerase D-like 1 OS
 Peroxisomal multifunctional enzyme type 2 OS
 Phosphatidylinositol-3-phosphatase SAC1 OS
 Piccolo (presynaptic cytomatrix protein) OS
 Plastin 1 OS
 Poly(A) binding protein__ cytoplasmic 4 OS
 Poly(rC) binding protein 2 OS
 Potassium channel subfamily K member 18 OS
 Potassium-transporting ATPase alpha chain 1 OS
 Potassium-transporting ATPase alpha chain 2 OS
 Proteasome 26S subunit__ ATPase 6 OS
 Protein arginine methyltransferase 1 OS
 Protein arginine N-methyltransferase 1 OS
 Protein phosphatase 3 catalytic subunit alpha OS
 Protein piccolo OS
 Purine nucleoside phosphorylase OS
 Pyruvate kinase PKM OS
 Ras-related protein Rab-3 OS
 Reticulon-3 OS
 Reversed Sequence 15407
 Reversed Sequence 16130

Reversed Sequence 16199
Reversed Sequence 24853
Ribosomal protein S3 OS
RING-type E3 ubiquitin transferase OS
SAC1 like phosphatidylinositol phosphatase OS
Septin-type G domain-containing protein OS
Serine (Or cysteine) peptidase inhibitor__ clade B__ member 6a OS
Serine (or cysteine) peptidase inhibitor__ clade B__ member 6b OS
Serp family B member 6A OS
Serpine1 mRNA binding protein 1 OS
SERPINE1 mRNA-binding protein 1 OS
Similar to Histidine triad nucleotide-binding protein 1 (Adenosine 5-monophosphoramidase)
(Protein kinase C inhibitor 1) (Protein kinase C-interacting protein 1) (PKCI-1) OS
Small ribosomal subunit protein eS1 OS
Small ribosomal subunit protein uS3 OS
Small ribosomal subunit protein uS8 OS
Sodium/potassium-transporting ATPase subunit alpha-4 OS
Solute carrier family 9 member C1 OS
Sortilin 1 OS
Sortilin OS
Sphingomyelin phosphodiesterase OS
Splicing factor U2AF subunit OS
Squalene synthase OS
Succinyl-CoA:3-ketoacid coenzyme A transferase 1__ mitochondrial OS
Succinyl-CoA:3-ketoacid-coenzyme A transferase OS
Sulfotransferase OS
Testin OS
Tetratricopeptide repeat domain 3 OS
Threonine-tRNA ligase 1__ cytoplasmic OS
threonine-tRNA ligase OS
Transcription intermediary factor 1-beta OS

x

Tripartite motif-containing 28 OS
Tyrosine-protein phosphatase non-receptor type 20 OS
V-type proton ATPase subunit H OS
Vasculin-like protein 1 OS
Vesicle-fusing ATPase OS
Vigilin OS

1.4.1.7 OGD y NaHS

Proteins identified only in the OGD and NaHS conditions

```
Venn_OGD_X_NaHS <- setdiff(intersect(PersulfuratedOGD, PersulfuratedNaHS), PersulfuratedControl)
Lista_OGD_X_NaHS <- setdiff(intersect(PersulfuratedOGD_protein.accession, PersulfuratedNaHS_protein.accession), PersulfuratedControl_protein.accession)
knitr::kable(sort(Venn_OGD_X_NaHS))
```

x

Aconitate hydratase__ mitochondrial OS
Adaptor related protein complex 1 subunit beta 1 OS
Aldose reductase-related protein 1-like OS
AP complex subunit beta OS
AP-1 complex subunit beta-1 OS
AP-2 complex subunit beta OS
ATP-dependent RNA helicase DDX1 OS
Bifunctional phosphoribosylaminoimidazole carboxylase/phosphoribosylaminoimidazole succinocarboxamide synthetase OS
Bifunctional purine biosynthesis protein ATIC OS
Calcium-transporting ATPase OS
Catenin beta 1 OS
Ceruleplasmin OS
Copine 6 OS
Cyclin-dependent kinase 14 OS
Cytosol aminopeptidase OS
Fc fragment of IgG binding protein-like 1 OS

Glyceraldehyde-3-phosphate dehydrogenase pseudogene OS
Guanine nucleotide-binding protein G(o) subunit alpha OS
Guanine nucleotide-binding protein G(s) subunit alpha isoforms short OS
Heat shock 70 kDa protein 1-like OS
Heat shock protein family A (Hsp70) member 4 like OS
hexokinase OS
Hydin__ axonemal central pair apparatus protein OS
Isoform 2 of AP-2 complex subunit beta OS
Isoform 2 of Cytosol aminopeptidase OS
Isoform 2 of Sarcoplasmic/endoplasmic reticulum calcium ATPase 2 OS
Isoform B of AP-1 complex subunit beta-1 OS
Keratin 84 OS
Keratin__ type I cytoskeletal 42 OS
Keratin__ type II cytoskeletal 6A-like OS
Keratin__ type II cytoskeletal 72 OS
Large ribosomal subunit protein uL10 OS
Malate dehydrogenase OS
Phosphoglycerate mutase 2 OS
Probable ATP-dependent RNA helicase DDX5 OS
Proteasome 20S subunit alpha 6 OS
Proteasome subunit alpha type-6 OS
Reticulon OS
Reticulon-4 OS
Reversed Sequence 21569
Sarcoplasmic/endoplasmic reticulum calcium ATPase 2 OS
Serine/threonine-protein phosphatase OS
Sodium/potassium-transporting ATPase subunit alpha-1 OS
Sodium/potassium-transporting ATPase subunit alpha-2 OS
T-complex protein 1 subunit alpha OS
Tubulin alpha-1A chain OS
Tubulin alpha-1B chain OS

x

Tubulin alpha-1C chain OS

Tubulin alpha-3 chain OS

Tubulin alpha-8 chain OS

1.5 Análisis Gene Ontology (GO)

El análisis GO comenzó con la extracción del identificador Uniprot vinculado a la identificación del péptido con la señal de DCP-Bio. Luego, este identificador se tradujo a ENSEMBL utilizando la biblioteca “biomaRt”. Para cada condición, se llevó a cabo el análisis GO utilizando herramientas como “clusterProfiler”, “AnnotationDbi” y la base de datos de Rattus norvegicus como fondo (“org.Rn.eg.db”).

```
# Function to perform GO enrichment and generate barplot
enrich_and_plot <- function(gene_list, org_db, key_type, ontology, show_category, axis_text_size)
  # Perform GO Enrichment
  enrichment_result <- enrichGO(
    gene = gene_list,
    OrgDb = org_db,
    keyType = key_type,
    ont = ontology
  )

  # Generate barplot with axis text customization
  plot <- barplot(enrichment_result, showCategory = show_category) +
    theme(
      axis.text.x = element_text(size = axis_text_size, angle = 45, hjust = 1),
      axis.text.y = element_text(size = axis_text_size),
      axis.title = element_text(size = axis_text_size + 2),
      plot.title = element_text(size = axis_text_size + 4, face = "bold"),
      legend.text = element_text(size = axis_text_size + 1)
    )
```

```

    return(plot)
}

# Function to save plot as PNG
save_plot <- function(plot, output_file, res = 250, width = 1400, height = 1800) {
  png(output_file, res = res, width = width, height = height)
  print(plot)
  dev.off()
}

# Define global parameters
org_db <- "org.Rn.eg.db"
key_type <- "UNIPROT"
ontology <- "BP"
show_category <- 15
axis_text_size <- 8

# Define gene lists
gene_lists <- list(
  Control = Lista_Control,
  Control_NaHS = Lista_Control_X_NaHS,
  Control_OGD = Lista_Control_X_OGD,
  Control_OGD_NaHS = Lista_Control_X_OGD_NaHS,
  NaHS = Lista_NaHS,
  OGD = Lista_OGD,
  OGD_NaHS = Lista_OGD_X_NaHS
)

# Directory to save plots
output_dir <- "plots/"
if (!dir.exists(output_dir)) dir.create(output_dir)

```

```

# Process all gene lists and save plots
plots <- lapply(names(gene_lists), function(name) {
  cat("Processing:", name, "\n")

  # Generate plot
  plot <- enrich_and_plot(
    gene_list = gene_lists[[name]],
    org_db = org_db,
    key_type = key_type,
    ontology = ontology,
    show_category = show_category,
    axis_text_size = axis_text_size
  )

  # Save plot
  output_file <- paste0(output_dir, "GO_", name, "_Plot.png")
  save_plot(plot, output_file)
})

```

```

## Processing: Control
## Processing: Control_NaHS
## Processing: Control_OGD
## Processing: Control_OGD_NaHS
## Processing: NaHS
## Processing: OGD
## Processing: OGD_NaHS

```